

T3 - Invariant Subspaces and Diagonalization

Definition (Invariant Subspace)

Let V be a vector space and $f : V \rightarrow V$ a linear map.

A subspace $W \subseteq V$ is said to be **f -invariant** if

$$f(W) \subseteq W.$$

In this case, the restriction

$$f|_W : W \rightarrow W$$

is a well-defined linear map.

Proposition (Basic Examples)

The following subspaces are always f -invariant:

- 0
- V
- $\ker(f)$
- $\operatorname{Im}(f)$

Proof

- If $v \in \ker(f)$, then $f(v) = 0 \in \ker(f)$.
- If $v = f(u) \in \operatorname{Im}(f)$, then $f(v) = f(f(u)) \in \operatorname{Im}(f)$.

The other cases are immediate.

Remark

Invariant subspaces need not be nontrivial.

Example (Rotation)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be rotation by an angle $\neq 0, \pi$.

Then no nontrivial subspace of $\mathbb{R}^2$ is f -invariant.

Reason

Every 1-dimensional subspace (line through the origin) is rotated to a different line.

Invariant Subspaces of the Differentiation Operator

Setting

Let

$$D : \mathbb{R}_4[X] \rightarrow \mathbb{R}_4[X], \quad D(p) = p'.$$

Proposition

For each $i = 0, 1, 2, 3, 4$, the space $\mathbb{R}_i[X]$ is D -invariant.

Proof

If $\deg p \leq i$, then

$$\deg D(p) \leq i - 1 \leq i,$$

hence $D(p) \in \mathbb{R}_i[X]$.

Example (Non-invariant Subspace)

Let

$$S = \langle x^4, x^3 \rangle.$$

Then S is not D -invariant because

$$D(x^4) = 4x^3 \in S, \quad D(x^3) = 3x^2 \notin S.$$

Proposition (Classification of D -Invariant Subspaces)

Let $W \subseteq \mathbb{R}_4[X]$ be a D -invariant subspace.

Then there exists $d \in 0, 1, 2, 3, 4$ such that

$$W = \mathbb{R}_d[X].$$

Proof

Let d be the maximal degree of a polynomial in W . Then $W \subseteq \mathbb{R}_d[X]$.

Choose $p \in W$ with $\deg p = d$. Since W is D -invariant,

$$p, D(p), D^2(p), \dots, D^d(p) \in W.$$

These polynomials have distinct degrees:

$$d, d-1, \dots, 0.$$

Hence they are linearly independent, so

$$\dim W \geq d+1.$$

But $\dim \mathbb{R}_d[X] = d+1$, hence

$$W = \mathbb{R}_d[X].$$

Conclusion

The only D -invariant subspaces are:

$$0, \mathbb{R}_0[X], \mathbb{R}_1[X], \dots, \mathbb{R}_4[X].$$

One-Dimensional Invariant Subspaces

Proposition

Let $f : V \rightarrow V$ be linear.

A subspace W with $\dim W = 1$ is f -invariant if and only if it is generated by an eigenvector of f .

Proof

Let $W = u$, $u \neq 0$.

- If W is f -invariant, then $f(u) \in u$, so

$$f(u) = \lambda u.$$

Thus u is an eigenvector.

- Conversely, if u is an eigenvector, then u is invariant.
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Example

Let

$$[f]_E = \begin{bmatrix} 3 & 0 & 0 & 1 & 3 & 0 & 0 & 0 & -2 \end{bmatrix}.$$

Then

$$\chi_f = (X - 3)^2(X + 2).$$

Compute eigenspaces:

$${}_3(f) = e_2, \quad {}_{-2}(f) = e_3.$$

Conclusion

The only 1-dimensional invariant subspaces are:

$$e_2, \quad e_3.$$

Invariant Complements

Definition

Let $W \subseteq V$ be f -invariant.

We say that W has an **f -invariant complement** if there exists $W' \subseteq V$ such that:

- W' is f -invariant
 - $V = W \oplus W'$
-

Example

For the previous map,

$$\mathbb{R}^3 = e_1, e_2 \oplus e_3.$$

Moreover,

$$f(e_1), f(e_2) \in e_1, e_2,$$

so e_1, e_2 is f -invariant.

Thus e_3 admits an invariant complement.

Proposition

e_2 does **not** admit an f -invariant complement.

Proof

Assume W is an invariant complement:

$$e_2 \oplus W = \mathbb{R}^3.$$

Then, in a suitable basis,

$$[f]_B = \begin{bmatrix} 3 & 0 & 0 & [f|_W] \end{bmatrix}.$$

Thus,

$$\chi_f = (X - 3)\chi_{f|_W}.$$

But

$$\chi_f = (X - 3)^2(X + 2),$$

so

$$\chi_{f|_W} = (X - 3)(X + 2).$$

Hence $f|_W$ is diagonalizable (distinct roots), implying f is diagonalizable.

This contradicts:

$$\dim_3(f) = 1 < 2.$$

Conclusion

Not every invariant subspace admits an invariant complement.

A 4-Dimensional Example Without Eigenvectors

Setting

Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ with

$$\chi_f = (X^2 + 1)^2.$$

Proposition

There are no 1-dimensional f -invariant subspaces.

Proof

A 1-dimensional invariant subspace corresponds to a real eigenvector.

But χ_f has no real roots.

Proposition

$\ker(f^2 + \text{id})$ is a 2-dimensional f -invariant subspace.

Proof

For any polynomial p , $\ker(p(f))$ is f -invariant.

Proposition (Uniqueness in Dimension 2)

There is a unique 2-dimensional invariant subspace.

Proof Sketch

If W' is another one, then $W \cap W'$ is invariant.

Possible dimensions: 0 or 2 (no dimension 1 invariant subspace).

Thus $W' = W$.

Proposition

There are no 3-dimensional invariant subspaces.

Proof

If $\dim W = 3$, then $\chi_{f|_W}$ has degree 3 and real coefficients.

Hence it has a real root.

But all roots must come from $\chi_f = (X^2 + 1)^2$, which has none in $\mathbb{R}$.

Contradiction.

Final Conclusion

All invariant subspaces of this f are:

$$0, \quad \ker(f^2 + \text{id}), \quad \mathbb{R}^4.$$

Summary

- Invariant subspaces encode internal structure of a linear operator.
- 1-dimensional invariant subspaces correspond exactly to eigenvectors.
- Invariant complements need not exist.
- Polynomial operators (like differentiation) yield rigid invariant structures.
- Absence of real eigenvalues severely restricts invariant subspaces.
- Dimension arguments combined with characteristic polynomials are the main tool for classification.